

Multiple Watermarking Scheme Using Adaptive Phase Shift Keying Technique

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Abstract

In this paper, a novel multiple-watermarking scheme using adaptive phase shift keying (APSK) modulation with amplitude boost (AB) strategy is proposed. AB is hired to increase the robustness. With proper combination of AB, the proposed scheme achieves superior performance in terms of robustness and imperceptibility.

In order to demonstrate the effectiveness of the proposed scheme, simulations under various conditions were conducted. The empirical results showed that APSK can sustain most common attacks including JPEG compression, rotating, resizing, cropping, painting, noising and blurring etc.

Keyword: Adaptive Phase Shift Keying (APSK), Discrete Fourier Transform (DFT), Amplitude Boost (AB), Joint Photographic Experts Group (JPEG).

1. Introduction

Recently, digital watermarking has been proposed as one solution of the problem of intellectual properties protecting [1-5].

Several methods in the transform domain is to hide watermarks in the discrete Fourier transform (DFT) coefficients of the host image. Ruanaidh et al. [6] presented a phase-based method in the DFT domain and used an optimal detector for watermark recovery. Based on the Fourier-Mellin Transform Ruanaidh and Pun [7] presented a watermarking scheme that achieves rotation, scale and translation (RST) invariant. The scheme achieves robustness while sustains the RST attacks. Premaratne and Ko [8] proposed a new concept for embedding and detecting the watermark in the Discrete Fourier Transform. Since the embedding is independent of the image content, speedy embedding highly suitable for video streams can be achieved. Solachidis and Pitas [9] proposed a watermarking scheme which embeds a circularly symmetric watermark on a ring in the 2D DFT domain. The circularly symmetric watermark was used to solve the rotation invariance problem in the watermark detection in which a correlation operation was used.

Akira et al. [13] based on the phase modulation

of the original signal is realized by means of a time-varying all-pass filter to accomplish the blind detection which is required in detecting the copy control information. In the scheme, the watermark is assigned to the inter-channel phase difference between a stereo audio signal by using frequency shift keying. In the same time, the copyright management information and the fingerprint are embedded into both channels by using phase shift keying of different frequency components. Consequently, these three kinds of information are embedded into a single time frame simultaneously.

In a previous paper [10], we proposed a DFT domain watermarking scheme using the phase shift keying (PSK). In the scheme, high-variance block selection (HVBS) is employed to improve imperceptibility. The PSK embedding is employed due to its superior noise immunity. In the PSK embedding, the watermark information is embedded in the phase part of the host image. Thus, the threshold effect in which the quality of the recovered watermark plunges when the amplitudes of the DFT coefficients used for embedding the secret bits are below a threshold value may occur. In this paper, we proposed a DFT domain watermarking scheme using the adaptive phase shift keying (APSK) modulation with amplitude boost (AB). AB is hired to increase the robustness. We demonstrated that by properly combining AB robustness can be enhanced without sacrificing imperceptibility. Meanwhile, in our scheme neither the original host image nor the original watermark is required during the watermark detection process.

The remainder of this paper is organized as follows. In section 2, the proposed concealing algorithm is presented. The watermark extracting process is presented in section 3. Empirical results are presented in section 4. Finally, section 5 concludes this paper.

2. Concealing Algorithm

A robust watermarking scheme must survive all kinds of attacks, and at the same time sustain the virtual quality of the host image when the watermark is concealed. Besides, security is also an important factor required by a watermarking scheme.

The overall concealing process of our proposed scheme is shown in Fig. 1. The watermark $W_1 \sim W_4$ is first transformed to H by chaotic mechanism (CM) using a pseudo random sequence (PN_3) generated by a private key to enhance the security respectively. It is then converted into a binary random sequence $\{m\}$ by rearrange to series bits (RSB) step. On the other hand, the host image X is transformed to Z by DFT. Then a block is selected from Z using the PN_2 sequence for secret data embedding. Two DFT coefficients $ae^{j\phi}$ and $ae^{-j\phi}$ which from a complex conjugate pair are selected from the selected block for the following amplitude boost (AB) process in which a is boosted to a' to combat attacks. In the APSK modulation, ϕ is modulated by m into ϕ' according its amplitude. After AB and APSK, $a'e^{j\phi'}$ and $a'e^{-j\phi'}$ in which ϕ' contains the secret information are then embedded into Z at the selected block and coefficient pair locations by watermark bits embedding (WBE). The above embedding process is repeated for each secret bit in $\{m\}$ and its corresponding block. The resultant image Z' after embedding is then inversely transformed to obtain the watermarked image Y . The details of each block of our embedding algorithm are presented in the following.

2.1 Chaotic Mechanism

In data hiding, we want the data to be chaotic because a chaotic data is not easy to be stolen or attacked. Therefore, it needs a chaotic mechanism to hash the output streams $\{m_c\}$. Once the output streams $\{m_c\}$ are chaotic, it is not easy to be attacked, and ensures better security. In this paper, a fast pseudorandom number traversing method is used as the chaotic mechanism to permute the output streams $\{m_c\}$. Fig. 2 shows the images transformed by chaotic mechanism.

2.2 Amplitude Boost

The AB strategy is to boost the amplitude of a selected DFT coefficient as shown in Fig. 3 to a threshold value th when its value is below th so that the phase distortion under attacks can be keep below a certain level. In other words, after AB the amplitude of all the DFT coefficients used for embedding the secret bits are all above th . That is,

$$a' = \begin{cases} th & , \text{ if } a \leq th \\ a & , \text{ if } a > th \end{cases} \quad (1)$$

2.3 APSK Embedding

2.3.1 Phase Modulation

In the phase modulation, the phase ϕ is modified into ϕ' according the embedded secret bit. Fig. 3 shows the signal constellation of the phase modulation for $s=1$, $s=1/2$ and $s=1/4$. Fig. 3(a) shows the two embedding location, the phase is 90 degree for secret bits='0' and the phase is 270 degree for secret bits='1'. The Fig. 3(b) is the situation for

divided the phase into two parts, $|\phi| \leq \frac{\pi}{2}$ and

$|\phi - \pi| \leq \frac{\pi}{2}$. In this case, the distortion of the phase

will be the half of the Fig. 3(a). Another case, a less distortion method is divided the phase into four part as the shown in Fig. 3(c).

2.3.2 Adaptive Phase Modulation Algorithm

Adaptive phase modulation is a mechanism that its phase strength is modulated according the signal amplitude automatically. In our approach, four kinds of modulation strength was used for secret bit embedding. The details of the adaptive phase modulation were described in Fig. 4. In the APSK, a secret bit and a coefficient were the input parameters and the modified coefficient is its output result. At the beginning, amplitude a was obtained from the coefficient which is processed by frequency hopping. Successively we check the amplitude; if its value is less than the threshold S_{th1} then the AB mechanism is enabling. AB lift its amplitude to the level of the S_{th1} , the level could resist the noise attack. Continually, we classify the signal amplitude into four classes; give their different phase modulation strength. There are $s=1/2$, $s=1/4$, $s=1/6$ and $s=1/8$ corresponding to the region $R=4$, $R=8$, $R=12$ and $R=16$ respectively. Finally, a phase function with the parameter region R and secret bit m_c was used to compute the modulated angle. After replace the selected coefficients signal phase with the angle, the adaptive phase modulation was finished. The formulas of the APSK are describing in the Eqs. (2)-(3).

$$\angle \theta = \phi(R, m_c) \quad (2)$$

$$B(i, j) = a \cdot e^{j\theta} \quad (3)$$

3. Watermark Extraction Algorithm

Robustness, imperceptibility and security are three issues concerned in our watermarking scheme. In the embedding process, the DFT, the AB is used to enhance the robustness. The APSK is used to increase the imperceptibility. The CM and the set of bit sequences (PN_2, PN_3) are hired for the security reason. In the extracting process, the secret data must be extracted from the same positions where they are embedded. Therefore, the same set of bit sequences (PN_2, PN_3) used in the embedding process is used in the extracting process. Beside, the APSK demodulation and ICM are used in the extracting process to inverse the operations of APSK modulation and CM in the embedding process respectively.

The block diagram of the recovering process is shown in Fig. 5. The watermarked image $\hat{Y}$ is transformed to $\hat{Z}$ by DFT. The same PN sequence PN_2 used in the concealing process is used to select the embedded blocks from $\hat{Z}$ for the APSK demodulation. In the APSK demodulation, the secret bit $\hat{m}$ is extracted from the phase $\hat{\phi}$ of the selected DFT coefficient for each selected block. After all the secret bits are extracted from the APSK demodulation, they are contracted by inverse rearrange to series bits (IRSB) and recover to the two-dimensional image $\hat{H}$. By passing $\hat{H}$ through the inverse chaotic mechanism (ICM), the recovered watermarks $\hat{W}_1 \sim \hat{W}_4$ are obtained.

3.1 APSK Demodulation

In the phase demodulation, the angle ϕ is processed to recover the embedded secret bit m . we adopt the APSK for phrase modulation in secret data embedding. Therefore an inverse action APSK demodulation is necessary for secret data extracting. Fig. 6 shows the flow chart of the APSK demodulation. In the process of APSK, the amplitude a of the selected coefficients $B(i, j)$ was used to check what region it belongs. After the region was obtained, pass it and phase information to the decision function. And then the secret data $\hat{m}$ was extracted. The decision function was given in Eq. (4).

$$\hat{m} = f(R, \phi(B(i, j))) \quad (4)$$

where the R denotes the region of the signal belongs, and $\phi(B(i, j))$ is the phase of the signal $B(i, j)$.

3.2 Inverse Chaotic Mechanism

In the secret data embedding process, a fast pseudorandom number traversing method is used as the chaotic mechanism to permute the output streams. The relation between the bit sequence after permutation and the bit sequence before permutation. In the secret data extracting, the inverse fast pseudorandom number traversing method is used as the inverse chaotic mechanism to re-permute the output streams .

4. Experimental Results

Imperceptibility is an important factor for watermarking. In this paper we employ the $PSNR$ to indicate the degree of transparency. The $PSNR$ of Y is given by

$$PSNR = 10 \log_{10} \frac{255^2}{\frac{1}{N \times N} \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} (X(i, j) - Y(i, j))^2} \quad (5)$$

Where $X(i, j)$ and $Y(i, j)$, are the gray values at (i, j) of the host image X and the watermarked image Y of size $n = N \times N$ respectively. The watermark similarity measurement is dependent on factors such as the knowledge of the experts, the experimental conditions, etc. Therefore a quantitative measurement is necessary to provide fair judgment of the extracted fidelity.

The images Boots (512×512) and Pepper (512×512) are used in simulation for demonstrating the performance of the proposed scheme. The logo image, National Chin-Yi Institute of technology (32×32) in Chinese, Department of electronic (32×32) in Chinese, VR LAB (32×32) and NCIT IEE (32×32) were used as the watermarks. The block size used is 8×8 . The values of parameters used to confined the range of phase in the simulations are: $S_{th1} = 280$, $S_{th2} = 310$, $S_{th3} = 450$ for Boots image and $S_{th1} = 300$, $S_{th2} = 330$, $S_{th3} = 350$ for the Baboon image.

Attacks include rotating, resizing, cropping, painting, blurring, noising and JPEG compression are used in the simulations to demonstrate the robustness of our scheme. For the rotating attack, the watermarked image is rotated 90 degree CW followed by 90 degree CCW before proceeding the extracting process. For the resizing attack, the

watermarked image is shrunk from 512×512 to 320×320 followed by expanding from 320×320 to 512×512 before proceeding the extracting process. For the cropping attack, the watermarked image is cropped 20% before proceeding the extracting process. For the painting attack, several paints are added to the watermarked image before proceeding the extracting process. For the blurring attack, the watermarked image is blurred by Gaussian convolution before proceeding the extracting process. For the noising attack, Gaussian noises are added into the test image before proceeding extracting process. Finally, the JPEG attack was used to compress the test image before proceeding extracting process.

In order to demonstrate cover image possesses excellent imperceptibility without noticeable degradation, we compare it with the test images.

Where the embedded image of Pepper has the PSNR is 37.48db and the embedded image of Boots with the PSNR=37.16 respectively.

On the other hand, the watermarked images under some attacks are shown in Fig.7 and Fig. 9. In Fig. 7 is the attacked watermarked images "Boots". Fig. 7(a) is the 90-degree rotated attack. Fig. 7(b) denotes the resized attack. Fig. 7(c) and Fig. 7(d) is the cropped and the painted attacks respectively. Finally, the noised and the blurred attack are shown in Fig. 7(e) and Fig. 7(f). Corresponding to the above attacks, the number of the error bits of the recovered watermarks (w_1, w_2, w_3, w_4) for Boots image is shown in table 1. Similarly, Fig. 9 is the attacks corresponding to Fig. 7 for Pepper image. And its error bits of the recovered watermarks (w_1, w_2, w_3, w_4) is shown in table 3.

Corresponding to the JPEG attacks, the table 2 and the table 4 show their comparison of the number error bits of the recovered watermarks for the Boots and Pepper image respectively. Furthermore, Meanwhile, the comparison of the number error bits of the recovered watermarks under noise added attacks for the Boots image and Pepper image is shown in Fig. 8 and Fig. 10 respectively. From the experiment results, we can see the qualities of the extracted watermarks are all good.

5. Conclusions

In this paper we present a robust multiple-watermarking scheme for still images using APSK with amplitude boost and nearest phase selection. The AB strategy is to boost the amplitude of a selected DFT coefficient to a threshold value th when its value is below th so that the phase

distortion under attacks can be keep below a certain level. In the APSK embedding, we are according the amplitude of the selected coefficient to distinguish several ranges that is the watermark embedding strength. Different amplitude selects different embedding strength dynamically. For APSK reason, the watermarked image decreases the degradation and improves imperceptibility. Empirical results show that the proposed scheme can sustain attacks like JPEG lossy compression, rotation, resizing, cropping, painting, noising and blurring.

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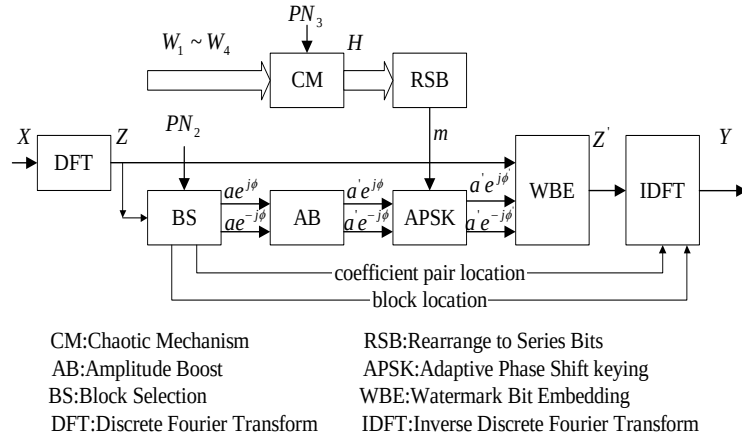


Fig. 1 The block diagram of the proposed embedding process



Fig. 2 The images transformed by chaotic mechanism.

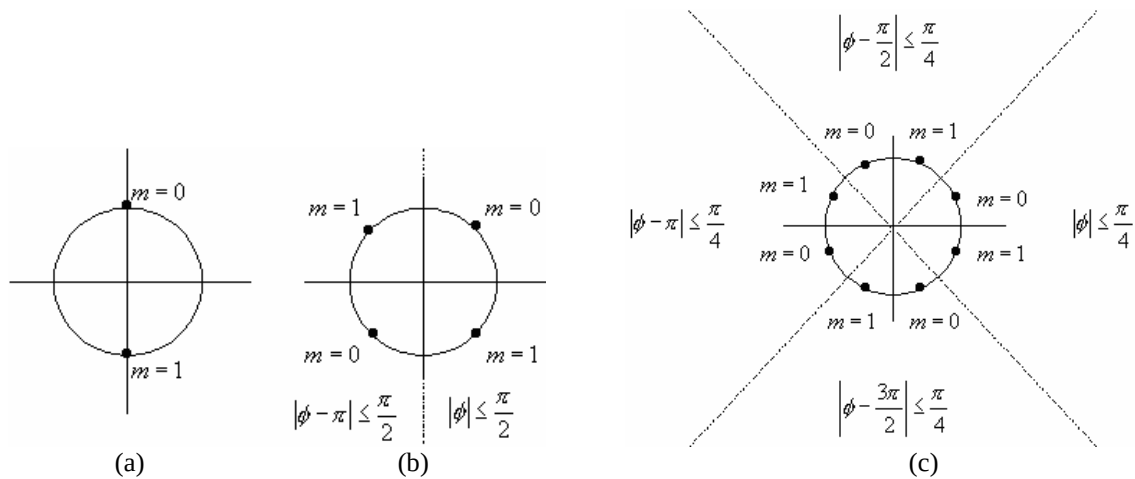


Fig. 3 The signal constellation of the PSK for (a) $s=1$, (b) $s=1/2$, and (c) $s=1/4$

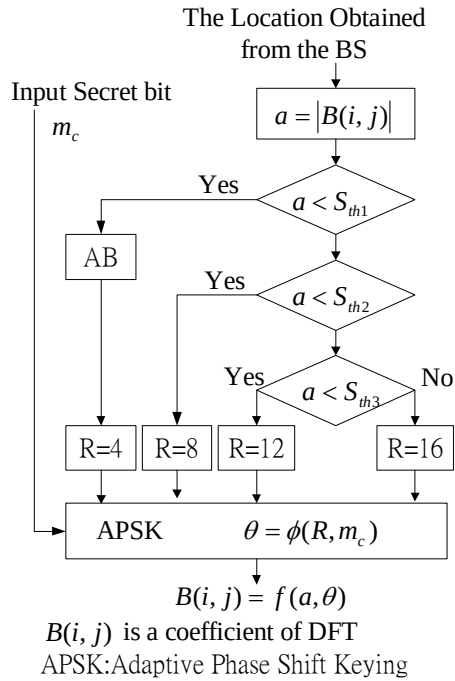


Fig. 4 The flow chart of the adaptive phase modulation

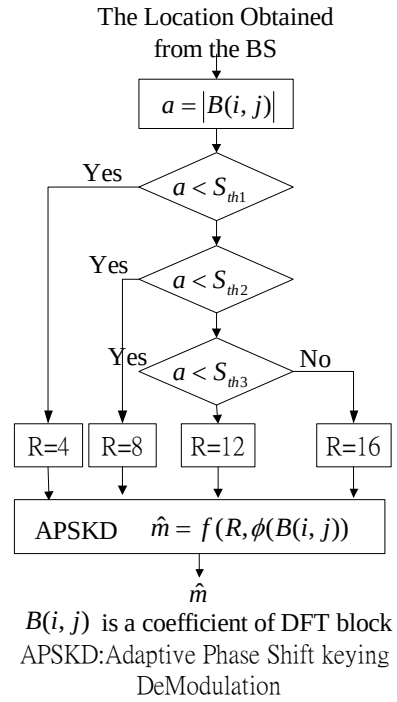


Fig. 6 The flow chart of the APSK demodulation

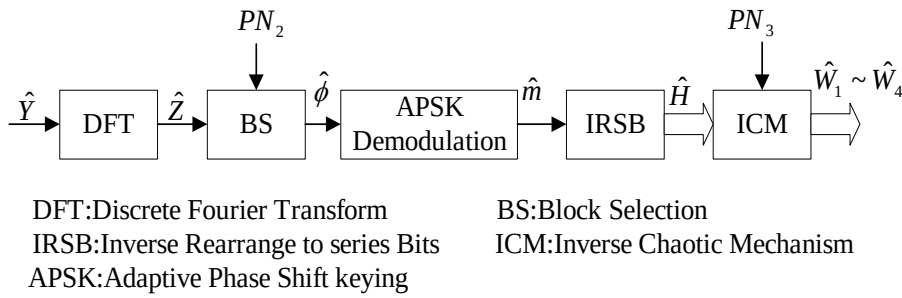


Fig. 5 The block diagram of watermark detection

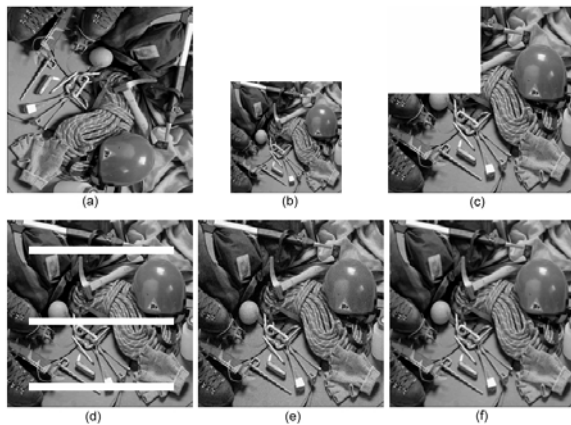


Fig.7 The attacked watermarked images (a - f), (a) the 90-degree rotated, (b) the resized, (c) the cropped, (d) the painted, (e) the noised, (f) the blurred.

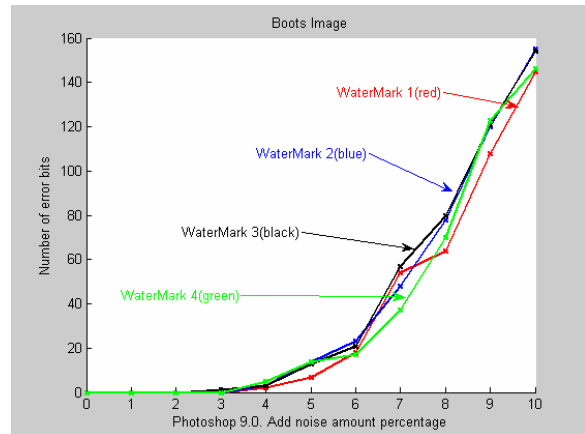


Fig. 8 The figure of the comparison of the number error bits of the recovered watermarks under noise added attacks for the Boots image.

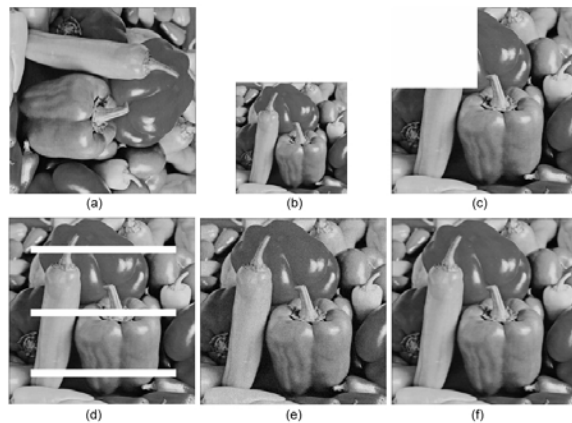


Fig.9 The attacked watermarked images (a - f), (a) the 90-degree rotated, (b) the resized, (c) the cropped, (d) the painted, (e) the noised, (f) the blurred.

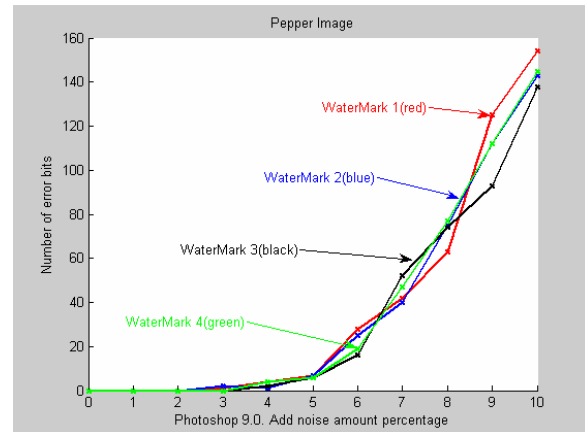


Fig. 10 The figure of the comparison of the number error bits of the recovered watermarks under noise added attacks for the Pepper image.

Table 1 Comparison of the number error bits of the recovered watermarks under corresponding Fig. 13 attacks for the Boots image

item	the rotated	the resized	the cropped	the painted	the noised	the blurred
W1 err. bits	0	38	75	80	54	56
W2 err. bits	0	34	72	71	48	51
W3 err. bits	0	52	89	71	57	73
W4 err. bits	0	40	78	53	37	57

Table 2 Comparison of the number error bits of the recovered watermarks under JPEG attacks for the Boots image.

quality	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7	Level 8	Level 9	Level 10	Level 11
W1 err. bits	90	6	1	0	0	0	0	0	0	0
W2 err. bits	82	5	0	0	0	0	0	0	0	0
W3 err. bits	80	3	0	0	0	0	0	0	0	0
W4 err. bits	81	5	1	0	0	0	0	0	0	0

Table 3 Comparison of the number error bits of the recovered watermarks under corresponding Fig. 16 attacks for the Pepper image

item	the rotated	the resized	the cropped	the painted	the noised	the blurred
W1						
W2						
W3						
W4						
W1 err. bits	0	14	87	75	42	18
W2 err. bits	0	25	88	56	40	30
W3 err. bits	0	12	93	63	52	16
W4 err. bits	0	15	81	74	47	17

Table 4 Comparison of the number error bits of the recovered watermarks under JPEG attacks for the Pepper image.

quality	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7	Level 8	Level 9	Level 10	Level 11
W1 err. bits	39	11	1	0	0	0	0	0	0	0
W2 err. bits	43	8	4	0	0	0	0	0	0	0
W3 err. bits	43	8	2	0	0	0	0	0	0	0
W4 err. bits	52	12	1	0	0	0	0	0	0	0